

iMVS API Definition

Imedtac

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Release Notes

Version	Date	Notes	Update staff
V1.0	11/09/2022	Draft	Alvin
V1.2	18/01/2023	Modify the default value of SPO2 to left-handed	Mars Yu
V1.3	11/07/2025	English version	Mars Yu
V1.4	12/05/2026	Updated auth and upload API	Ben Siu

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1. API

1.1. Authentication (provided by the hospital: query Patient from HIS)

Purpose	It is an optional API. When users try to login to iMVS by providing ID (via manual input, NFC or card reading), the ID will be sent to this API for authentication. If this API responded with Result="F", iMVS will reject the user. If this API is not implemented, all users are allowed to login to iMVS.		
URL	Provided by hospital		
Method	POST		
Parameter	Type	Required	Description
ID	String	Y	ID to authenticate

Callback data fields

Field ID	Parameter description	Type of data
Result	"T" = success, "F" = failure	string
Err	Error message, empty on success	string
ID	Original ID submitted	string
IdDisplay	Masked ID, e.g. A12****789. It will be used for display.	string
Payload	(optional) extra payload. It will be sent back to hospital via Vital Sign upload API	string

JSON Sample

Request body (JSON) (from iMVS to hospital)

```
{  
  "ID": "string"  
}
```

Response body (JSON) (from hospital to iMVS)

Success example:

```
{  
  "Result": "T",  
  "Err": "",  
  "ID": " A123456789",  
  "IdDisplay": " A12****789"  
}
```

Failure example:

```
{  
  "Result": "F",  
  "Err": "AUTH_FAILED",  
  "ID": "A123456789",  
  "IdDisplay": "A12****789"  
}
```

1.2. Vital sign upload

Purpose	Upload vital sign data after all measurements are done.
URL	Provided by hospital
Method	POST

The hospital returns the data field

Field ID	Parameter description	Type of data
success	Whether the setting is successful	Boolean
errorCode	Error Code	Int
message	Failure Message	String

Json Sample

```
{
  "success":true, "errorCode" : "", "message":""
}
```

Request body (from iMVS to hospital)

Field ID	Parameter description	Type of data
CHART_NO	ID (same as "ID" in auth API)	string
SAVE_DATETIME	Storage time of the data (measurement time) (yyyy-MM-dd HH:mm:ss)	string
UPLOAD_DATETIME	Upload date and time (yyyy-MM-dd HH:mm:ss)	string
STATION_NAME	Vital Sign Station name	string
Payload	Payload from auth API	string
SPO2	Blood oxygen	object
HR	Heart rate	object
Temp	Body temperature	object
Glucose	Blood glucose	object
NBP	Blood pressure	object
Height	Body height	object
Weight	Body weight	object

Measurement Parameters

2.2.1 SPO2 (Blood Oxygen)

Field ID	Parameter description	Type of data
Value	Measurement value	string
Unit	Unit (%)	string

2.2.2 HR (Heart Rate)

Field ID	Parameter description	Type of data
BP_Value	Heartbeat from blood pressure measurement	string
Unit	Unit (bmp)	string

2.2.3 Temp (Body temperature)

Field ID	Parameter description	Type of data
Value	Measurement value	string
Unit	Unit	string

2.2.4 Glucose (Blood glucose)

Field ID	Parameter description	Type of data
Value	Measurement value	string
Unit	Unit	string

2.2.5 NBP (Blood pressure)

Field ID	Parameter description	Type of data
SYS_Value	Systolic blood pressure value	string
DIA_Value	Diastolic blood pressure value	string
Unit	Unit	string

2.2.6 Height

Field ID	Parameter description	Type of data
Value	Measurement Value	string
Unit	Unit	string

2.2.7 Weight

Field ID	Parameter description	Type of data
Value	Measurement Value	string
Unit	Unit	string

JSON Sample

```
{
  "CHART_NO": "P103011",
  "SAVE_DATETIME": "2019/10/30 09:40:24",
  "UPLOAD_DATETIME": "2019/10/30 09:40:28",
  "STATION_NAME": "SICU",
  "NBP": {
    "SYS_Value": "120",
    "DIA_Value": "80",
    "Unit": "mmHg"
  },
  "SPO2": {
    "Value": "98",
    "Unit": "%"
  },
  "HR": {
    "BP_Value": "72",
    "Unit": "bpm"
  },
  "Temp": {
    "Value": "36.5",
    "Unit": "°C"
  },
  "Glucose": {
    "Value": "80",
    "Unit": "mg/dL"
  },
  "Weight": {
    "Value": "70.0",
    "Unit": "kg"
  },
  "Height": {
    "Value": "175.0",
    "Unit": "cm"
  }
}
```


}

}